

Week 3 ML Submission

I am sohail , in this week I have learnt about ensembling techniques like bagging,boosting,random forest , xgboost , adaboost and I have implemented these algorithms on a single dataset(metro traffic volume) and *analyzed which algorithm is performing better and why.*

Initially I have loaded all the libraries and the given dataset and have done some operations like checking shape , looking up for null values and similar basic operations.

I have found over 48k null values in holiday column so I replaced them with “NONE” word.also I found holiday ,weather_main, weather_description are object columns, so I used label encoding on those columns

feature engineering was performed on the date_time column by extracting useful features like hour,day,month, week number,and weekend information.Adding these new columns have improved model performance significantly.

Later I have looked for outliers in the dataset and found rain_1h and snow_1h columns have outliers so using iqr method I have removed them .

Using train test split method ,I have divided the entire data into 5 parts , used 4 /5th part for training and 1/5th for testing(test size=0.20).

finally I created 3 models using **RANDOM FORESTS , XGBOOST , ADABOOST** using sklearn libraries.

RESULTS:

Random Forest:

R2 Score :0.963
MAE : 224.43
RMSE : 382.57

ADABOOST:

R2 Score :0.819
MAE :629.70
RMSE : 844.33

XGBOOST:

R2 Score :0.970
MAE : 216.54
RMSE : 344.10

Conclusion: XGboost has performed better than other models with highest r2 sscore and lowest Mae and RMSE ,because XGBOOST uses boosting technique where each tree focuses on correcting the mistakes made by previous trees , so xgboost captured traffic patterns better than adaboost and random forests which resulted in higher accuracy and lower error